The Particle Model — Mark schemes

PAPERS A TO D · 30 MARKS EACH · TUTOR COPY

Marks in square brackets show how the total is split. Most marks on this topic are won or lost on language rather than knowledge: an answer that describes how something feels or looks scores nothing, while the same idea expressed in terms of arrangement, movement, spacing or forces scores in full. Two phrases are worth insisting on — "the forces between the particles", and "harder and more frequently". The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

1 Arrangement — liquid: touching but jumbled, no pattern; gas: far apart, random [6]
Movement — solid: vibrate about fixed positions; gas: fast and random, in all directions
Spacing — liquid: close, still touching; gas: far apart, large spaces

One mark per box. "Do not move" for a solid is wrong — solid particles vibrate.

2 (a) Melting [4]
(b) Condensing
(c) Boiling (or evaporating)
(d) Sublimation

3 (a) Liquid [2]
(b) Gas

4 Solid: circles touching in a regular pattern [1] [3]
Liquid: circles touching but jumbled, no pattern [1]
Gas: circles well spread out with clear space between them [1]

Drawing gaps between the particles of a solid or liquid loses that mark. Only gases have real space.

5 In a gas the particles are far apart with large spaces between them [1] [3]
so they can be pushed closer together [1]
In a liquid the particles are already touching, so there is almost no space left to remove [1]

6 Melting point 0 °C [1] [2]
Boiling point 100 °C [1]

7 The particles are held in fixed positions by strong forces between them [1] [2]
so they cannot move past one another and the shape cannot change [1]

8 A physical change means no new substance is made [1], and the change can be reversed [1] [3]
Evidence: steam can be condensed straight back into water, and the mass is unchanged [1]

9 20 g [1] [2]
The same particles are still present, just arranged differently, and none can leave a sealed container [1]

10 Any three, one mark each: no fixed shape; no fixed volume / fills its container; can be compressed; much less dense than the solid or liquid; particles move fast and randomly; exerts pressure on the container walls. [3]

Paper B — Medium

30 MARKS

1 (a) 80 °C [1]; the temperature stops rising and stays at that value while the substance melts [1] [5]
(b) From 4 min to 8 min, so 4 minutes [1]
(c) The energy going in is breaking the forces holding the particles in their fixed pattern [1], so the particles become free to slide past each other rather than speeding up [1]

Notice the readings rise 24 °C in the first two minutes and 22 °C in the last two — the heater was working just as hard during the flat section.

2 Liquid [1] [3]
20 °C is above its melting point of −114 °C, so it is not solid [1]
and below its boiling point of 78 °C, so it is not a gas [1]

Both comparisons are needed. One of them alone narrows it to two states, not one.

3 All the energy going in is used to break or weaken the forces between the particles [1] [3]
None of it is left over to make the particles move faster [1]
Temperature is a measure of how fast the particles move, so it does not rise [1]

4 (a) The reading does not change [1] [4]
(b) The same particles are still inside; boiling only spreads them out and speeds them up [1], and nothing can enter or leave a sealed flask [1]
(c) The reading would fall, because steam would escape from the flask [1]

5 (a) Diffusion [1] [4]
(b) The particles of ink and water are in constant random motion and collide with each other [1]; because the ink starts crowded in one place, random movement carries more particles outwards than back, until they are evenly spread [1]
(c) Warming the water (accept: a shallower container, or gentle warming) [1]

"It just spread out" describes the result rather than explaining it, and scores nothing in (b).

6 (a) Brownian motion [1] [3]
(b) Air particles, too small to see, are moving fast and randomly [1] and colliding unevenly with the much larger smoke specks, knocking them about [1]

7 In a warm room the particles have more energy [1] [3]
so they move faster [1]
and diffuse across the room more quickly [1]

8 Sublimation [1] [2]
Example: solid carbon dioxide (dry ice), or iodine when warmed [1]

9 In the fridge the air particles inside lose energy and move more slowly [1] [3]
so they hit the walls of the balloon less often and less hard [1]
The pressure inside falls, so the outside air pressure squeezes the balloon smaller [1]

Paper C — Hard

30 MARKS

1 One mark for each genuine comparison: [4]
Boiling happens only at the boiling point; evaporation happens at any temperature.
Boiling happens throughout the liquid; evaporation happens only at the surface.
Boiling is fast; evaporation is slow.
Boiling produces bubbles; evaporation does not.

Each mark needs both halves. "Evaporation is at the surface" on its own is half an answer.

2 The particles in the puddle have a range of speeds, not all the same [1] [3]
The fastest ones at the surface have enough energy to overcome the forces holding them in the liquid [1]
so they escape as a gas, and this can happen at any temperature [1]

3 Only the fastest particles have enough energy to escape [1] [3]
so the average speed of the particles left behind falls [1]
Temperature is a measure of average particle speed, so the remaining liquid cools [1]

4 Any three, one mark each: higher temperature; larger surface area; wind or a draught. [3] [4]
One explained [1] — e.g. a higher temperature means more particles have enough energy to escape from the surface.

5 The particles gain energy and move faster [1] [3]
so they hit the walls of the tin harder [1] and more frequently [1]

Harder and more often. Only one of the two is the most commonly dropped mark in the whole topic.

6 The pressure doubles, or increases [1] [3]
The same number of particles now occupies half the space, so they hit the walls more often [1]
More frequent collisions in the same area means a greater pressure [1]

The particles are not moving faster — the temperature is unchanged. Only the frequency of collisions has changed.

7 (a) Diffusion [1] [4]
(b) In the empty jar the bromine particles travel freely in straight lines with nothing in the way [1]; in air they collide constantly with air particles [1], so they are knocked in random directions and take a long, zig-zagging path to cross the jar [1]

The bromine particles are moving just as fast in both jars. It is the path length, not the speed, that differs.

8 Water particles escape from the surface of the wet clothes by evaporation [1] [3]
Wind carries those escaped particles away from just above the surface [1]
so fewer are recaptured by the cloth, and the washing loses particles faster overall [1]

9 When steam touches the skin it condenses back into liquid water [1] [3]
Condensing releases the large amount of energy that was needed to separate the particles when it boiled [1]
That extra energy is transferred to the skin on top of the energy from the water cooling, so the burn is worse [1]

This is the same 2260 kJ per kilogram that made the boiling plateau so long on the study page.

Paper D — Hard

30 MARKS

1 (a) 55 °C [1] [6]
(b) At 2 minutes it is a liquid (72 °C, above the freezing point) [1]; at 14 minutes it is a solid (30 °C, below it) [1]
(c) The particles are settling into fixed positions in a regular pattern [1]; as they do so, energy is released [1], and this released energy replaces the energy lost to the room, so the temperature holds steady until all of it has frozen [1]

Freezing releases energy — the mirror image of melting absorbing it. Students who describe freezing as "losing energy so it cools" have missed the point of the flat section.

2 (a) By diffusion [1]; the gas particles are in constant random motion and spread out from where they are concentrated towards the empty space in the tube [1] [5]
(b) The ring forms where the two gases meet. Ammonia particles are lighter, so at the same temperature they travel faster [1] and cover more of the tube before the two meet, putting the meeting point nearer the hydrogen chloride end [1]
(c) The ring would form in about the same place, but sooner — both gases speed up, so the race is still won by the same margin [1]

Part (c) is the discriminator. The instinct is to say the ring moves, because "faster" feels as though it should change the position.

3 Award one mark for each of these, up to six: [6]
Below 0 °C the energy makes the particles vibrate faster and the temperature rises.
At 0 °C the temperature stops rising; the energy breaks the forces holding particles in the lattice.
The particles become free to slide past each other — the ice melts.
From 0 to 100 °C the energy again goes into speed, and the temperature rises.
At 100 °C the temperature stops again; the energy separates the particles completely.
Above 100 °C the energy goes back into speed and the steam heats up; throughout, the particles themselves never change.

For full marks the answer must say what the energy is doing at each stage, not just what state the substance is in. Award at most four if both flat sections are described without explaining where the energy goes.

4 (a) Liquid [1]; 20 °C is above the melting point of −7 °C and below the boiling point of 58 °C [1] [4]
(b) Gas [1]; 80 °C is above the boiling point of 58 °C [1]

5 (a) The water has turned to steam and escaped into the room as a gas [1]; the particles have left the pan, taking their mass with them [1] [4]
(b) No particles have been destroyed — they have only moved somewhere else [1]; if the steam were collected and weighed, the total would be unchanged, so mass is still conserved [1]

The pan is an open system. Conservation of mass only looks broken when you weigh part of it.

6 (i) The mistake: particles do not change size, whatever the temperature [1] [5]
Correct version: the particles vibrate faster and further apart, so the spaces between them increase and the solid expands [2]
(ii) The mistake: the energy is still going in at the same rate, so a great deal is happening [1]
Correct version: the energy is breaking the forces between the particles rather than speeding them up, so the state changes while the temperature stays constant [1]

These are the two most common wrong sentences in the whole topic. If he can mark them in someone else's work, he is unlikely to write them in his own.